Proxmox: Storage, Backup, Capacity, and Naming

Purpose: Documents the virtualisation host's storage layout, the backup and recovery strategy, the capacity plan for the planned estate, and the lab-wide machine naming convention. Written so that a reviewer can see not just *what* was configured but *why*, and which controls it satisfies.

Host: `proxmox-01` (`192.168.10.6`), Proxmox VE 9.2.2, standalone node.

Status: Backup configured and verified 2026-08-30.

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1. Host overview

Resource	Value
CPU	8 cores/threads
RAM	~32 GiB
Storage	2 physical disks, 2.67 TiB total
Cluster	Standalone (no cluster)

RAM is the binding constraint for how many guests run at once; CPU and storage are abundant.

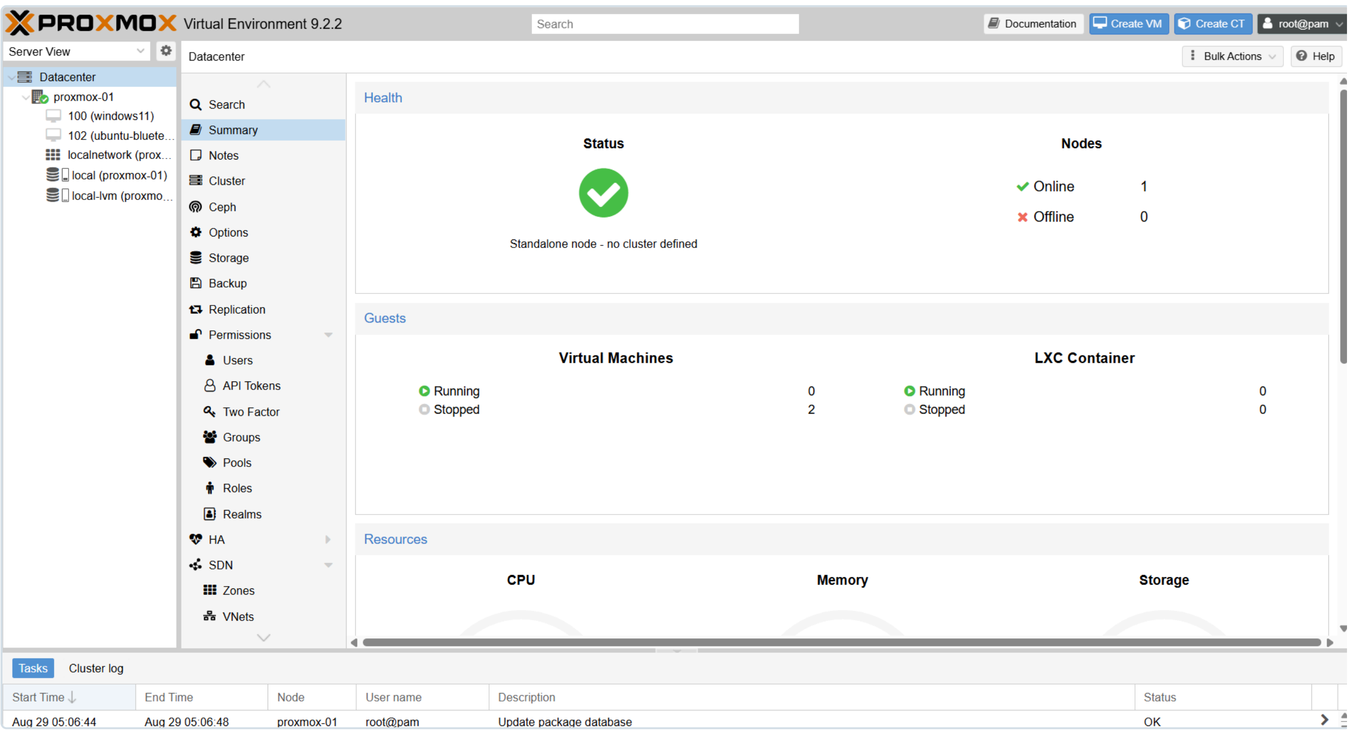

Figure 14.1: Datacenter summary: a healthy standalone node (`proxmox-01`), no cluster, with the current guests. This is the top-level health and inventory view.

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2. Storage layout

Disk	Size	Role
`/dev/sda`	3 TB	Proxmox OS + primary guest storage (`local` , `local-lvm`)
`/dev/sdb`	500 GB	**Backup target** (`backup` , ext4 Directory storage)

`sda` holds the two default install storages: `local` (ISO images, templates, backups directory) and `local-lvm` (VM/CT disk images). `sdb` was previously an OS disk (leftover EFI + ext4 partitions); it was wiped and repurposed as a dedicated backup store.

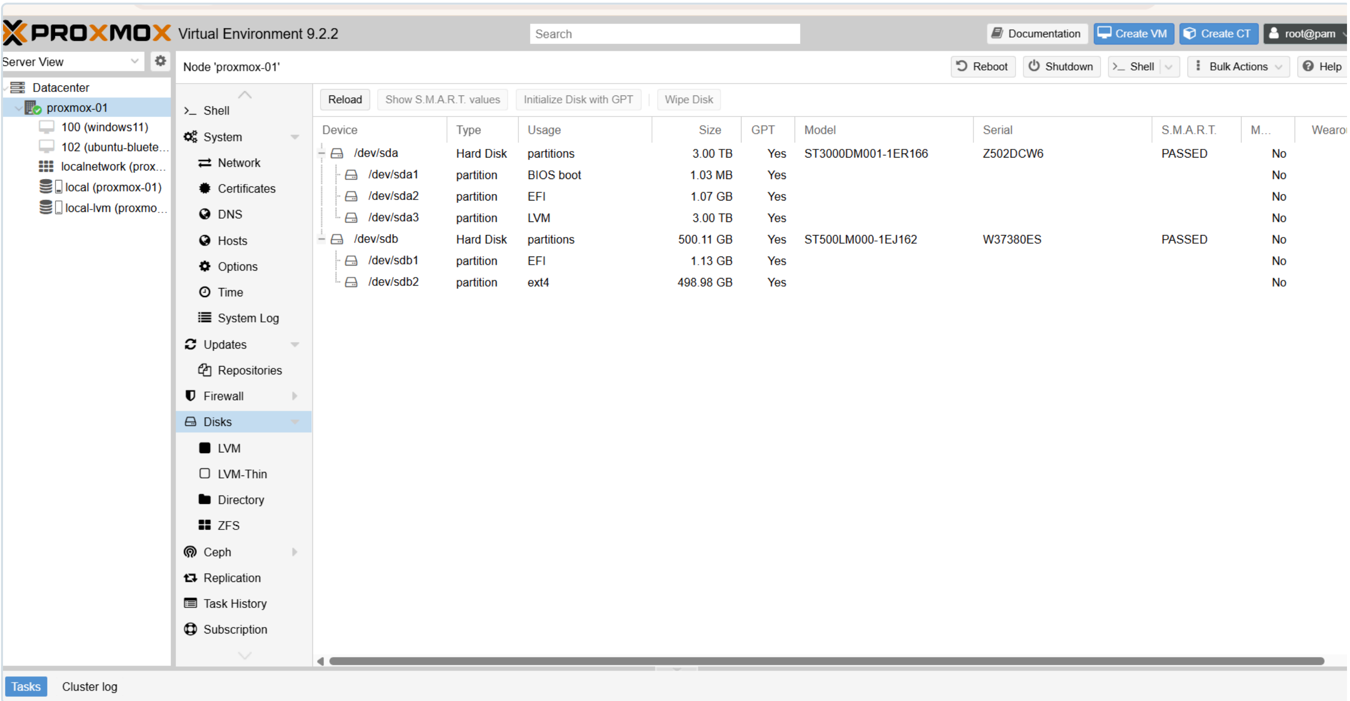

Figure 14.2: The two physical disks: `/dev/sda` (3 TB, Proxmox OS + LVM guest storage) and `/dev/sdb` (500 GB), before `sdb` was repurposed. Both report SMART "PASSED".

Why backups go on a separate disk

Keeping backups on a

different physical disk

from the running VMs means a failure or corruption of the primary disk does not take the backups with it. This is a basic tenet of recoverable design: the backup must survive the thing it is protecting against.

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3. Backup and recovery strategy

The principle

A running VM is not a backup of itself. Snapshots help with quick rollback, but a real backup is an independent, retained copy that survives disk loss, a bad change, or a corrupted guest. Two recent physical-machine failures in this lab (a server NIC, and the Wazuh host losing power) made the point concretely: **anything not backed up is one hardware fault away from a rebuild.**

What was configured (2026-08-30)

1. **Wiped** `/dev/sdb` and created a **Directory storage** named `backup` (ext4, mounted at `/mnt/pve/backup`), content type *VZDump backup file*.

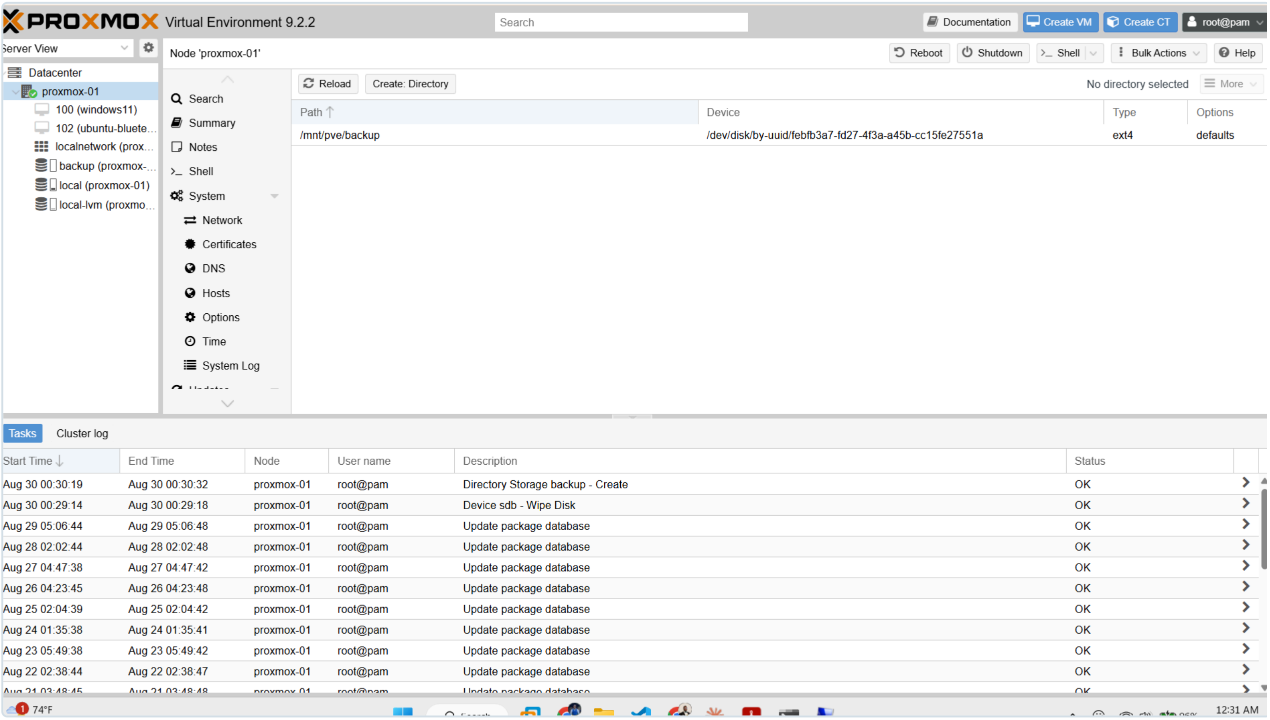

Figure 14.3: The `backup` Directory storage mounted at `/mnt/pve/backup` on the ext4 `sdb` disk. This is the dedicated backup target, separate from the disk that holds the running guests.

2. **Scheduled backup job**:

Setting	Value	Reason
Schedule	Daily 02:00	Off-hours, low activity
Selection	**All guests**	Auto-includes future VMs/CTs; no rule to forget
Mode	Snapshot	Backs up running guests with minimal disruption
Compression	ZSTD	Fast with good ratio
Retention	Keep last 3	Recent restore points without filling the disk
Target	`backup` (separate disk)	Survives primary-disk loss

3. **Verified** with an on-demand run: task completed `TASK OK`, backup files present under `backup` → Backups for both existing guests.

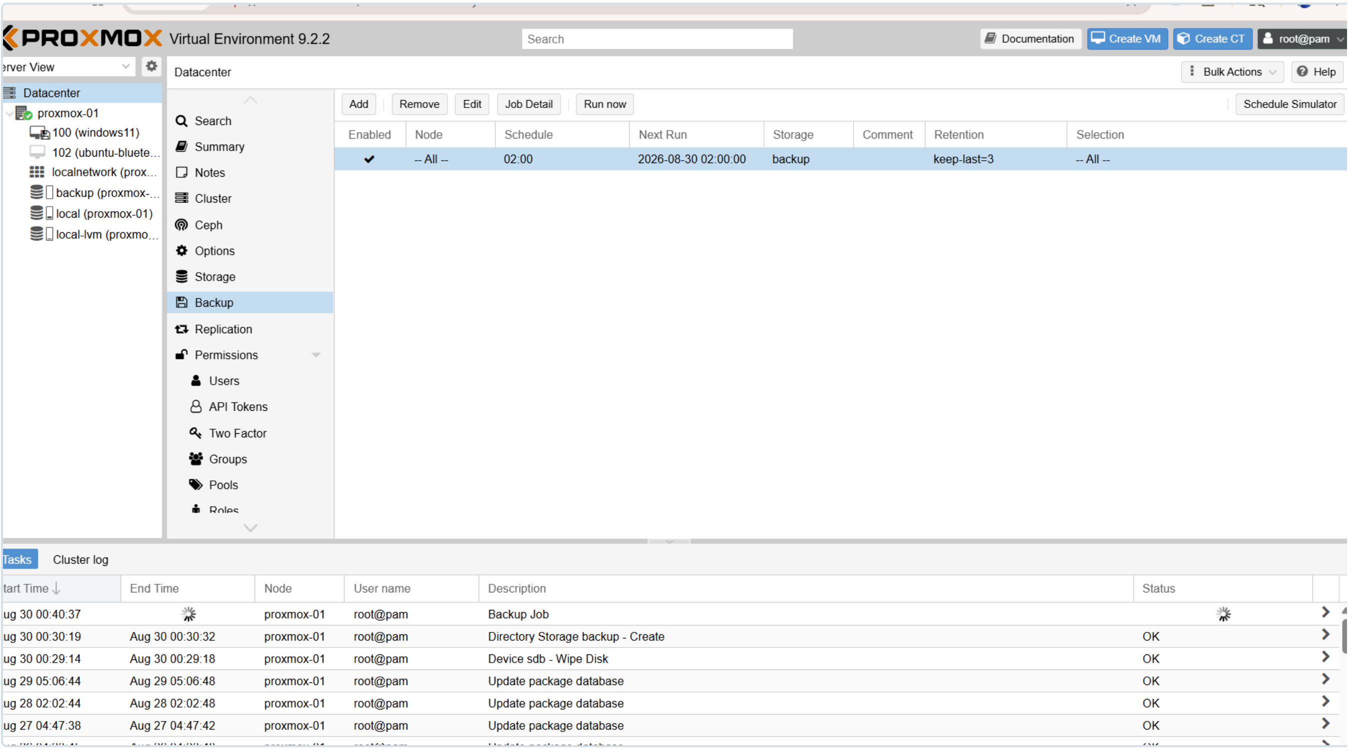

Figure 14.4: The backup job: enabled, all nodes, daily 02:00, target `backup`, retention keep-last=3, selection All. The running task at the bottom is the verification run.

What it protects, and what it does not

- **Protects**: all Proxmox VMs and containers (current and future). A lost or broken guest becomes a **restore**, not a rebuild.
- **Does not protect**: physical machines (e.g., the pfSense box, and a future physical DC02). Those need their own backup approach (for a Windows DC, the AD database is also protected by having a *second* DC, plus Windows Server Backup / system state).

Recovery

To restore a guest: Proxmox UI → `backup` storage → Backups → select the backup → Restore. Test restores periodically; a backup never verified by a restore is only a hope.

Future upgrade

Proxmox Backup Server (PBS) adds deduplication and incremental backups (far less space, faster). Not needed for a single node now; a Directory target is sufficient. Revisit if backup volume grows.

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4. Capacity plan

RAM is the limit (~32 GiB). The rule that keeps the lab within budget:

> **Linux services run as LXC containers; Windows runs as full VMs.**

LXC containers share the host kernel, a fraction of the RAM and disk of a VM. Windows cannot be a container, so it uses full VMs.

Guest	Type	Planned RAM
DC01 / DC02 (if virtual)	VM	2–4 GB
CA01	VM	2–4 GB
WKS01 / WKS02	VM	4 GB each
ANS01 (Ansible)	**LXC**	~1 GB
SIEM01 (Wazuh)	**LXC/VM**	~4 GB
MON01 (Grafana/Prometheus)	**LXC**	~2 GB
Kali	VM	~3 GB

All-running total is roughly 28 GB, within 32 GB, and guests can be powered on **on demand** rather than all at once. Verdict: the host comfortably carries the planned estate.

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5. Machine naming convention

A consistent, role-based naming scheme (adopted 2026-08-30). Names state the role, so any machine is identifiable from its name alone. This supports asset inventory and audit (NIST CM-8).

Role	Name(s)
Domain controllers	`DC01` , `DC02`
Certificate Authority	`CA01`
File / SQL servers	`FS01` , `SQL01`
Workstations	`WKS01` , `WKS02`
Wazuh / SIEM	`SIEM01`
Grafana / monitoring	`MON01`
Vault	`VAULT01`
Ansible controller	`ANS01`
Attack / offensive box	`KALI01`

Migration note: the first domain controller was originally named `SRV1` and was **renamed to `DC01` on 2026-08-30** to match this convention. The rename was change-controlled: `dcdiag /test:dns` was clean before and after, the IP `192.168.50.2` and the domain `ad.biira.online` were unaffected, and the Netlogon service was restarted to re-register the DC's SRV records under the new name. The firewall alias retains its original name `SRV1_DC` (it points to the IP, not the hostname). Existing guests `100 (windows11)` and `102 (ubuntu-blueteam)` will be renamed to their role names when rebuilt/repurposed.

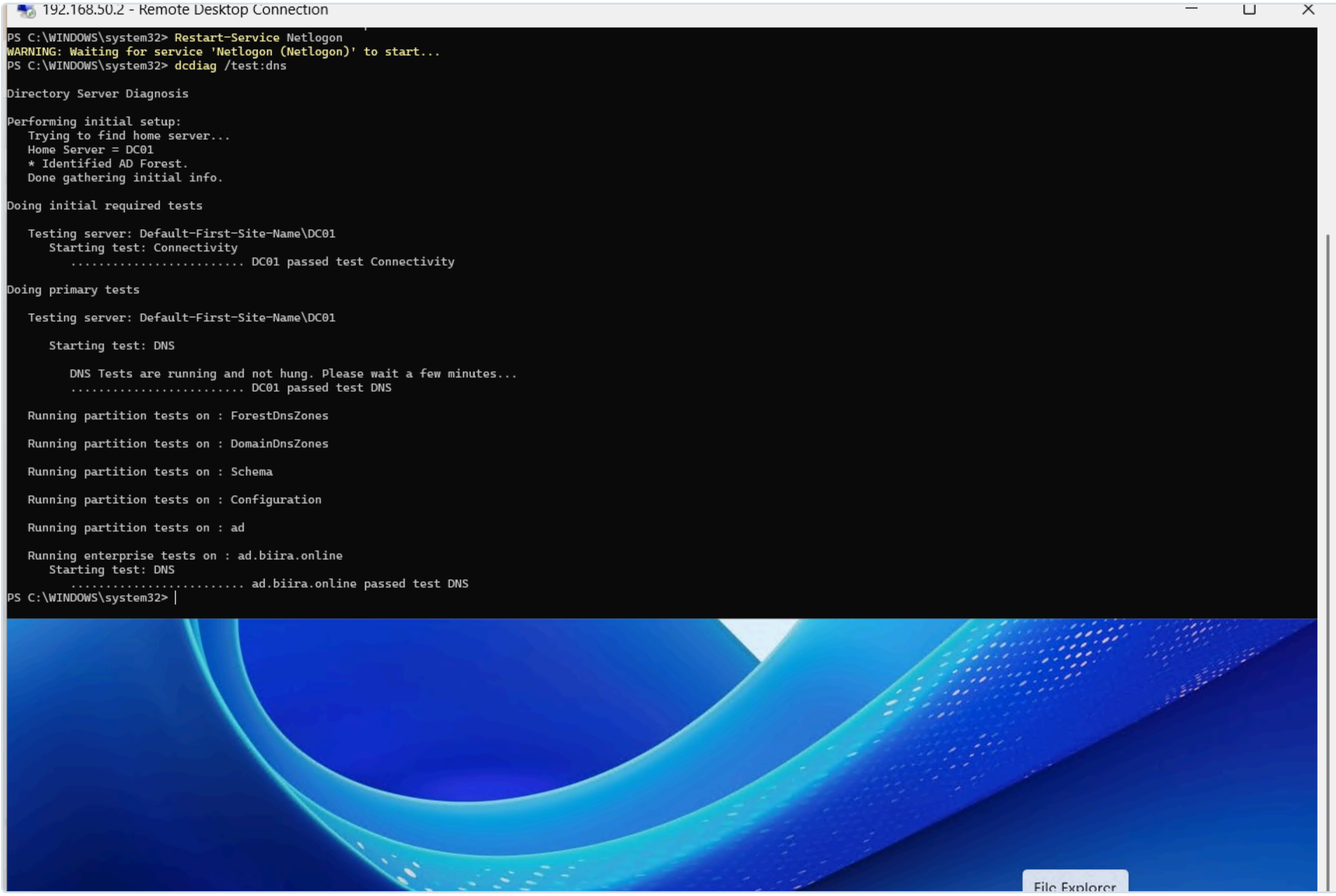

Figure 14.5: `dcdiag /test:dns` on `DC01` after the rename and Netlogon restart: Connectivity, DNS, all partition tests, and the enterprise `ad.biira.online` DNS test all pass. Evidence the rename left Active Directory healthy.

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6. Standards alignment

What we did	Control
Independent, retained, scheduled backups	**NIST SP 800-53 CP-9** (System Backup)
Documented restore procedure; periodic restore testing	**NIST CP-10** (System Recovery) / **CP-4** (Contingency testing)
Backups on separate disk from primary	Recoverability / defence against single-disk loss
Role-based naming for identifiable assets	**NIST CM-8** (System Component Inventory)
This document itself	**NIST CM-2/CM-6** (Baseline / documented configuration)

These are lab-appropriate implementations, not a claim of formal compliance.